

RESEARCH ARTICLE

Characteristics of pipestems from acid sulfate soils of Finland and Australia

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Abstract

This study describes iron (Fe) pipestems formed around and within root channels during the development of a homogeneous unripe sulfidic sediment into a ripe soil where redox- or pH-active elements segregate. Pipestems and adjacent soil material samples were characterised by chemical and mineralogical analyses and scanning electron microscopy (SEM) associated with energy dispersive spectroscopy (EDS). Pipestems occurred at 60–190 cm and at 30–130 cm in a young Finnish soil and a more mature Australian soil, respectively, while sulfidic materials were found below 170–180 cm. The pipestems consisted of mineral grains cemented together by Fe precipitate. There was an up to 24-fold enrichment of Fe and an up to 13-fold enrichment of sulfur (S) in the Finnish pipestems compared to the adjacent soil material, whereas the corresponding enrichment in the Australian soil was up to 27-fold for Fe and up to 8-fold for S. In the Australian pipestem matrices, the Fe concentration was as high as 40% compared to 14% in the Finnish ones. It was estimated that about 70 ton ha⁻¹ S had been mobilised from the sulfidic material in the Finnish soil at the depth of 50–150 cm. Part of S has leached out but a substantial amount remained in the soil constituting the stock of retained acidity. Almost all pipestems contained a new solid phase precipitated within the former cortex cells of the plant roots. In the Finnish samples, this precipitate consisted of jarosite and schwertmannite. After oxidative exhaustion of sulfidic material in the surrounding soil, these metastable minerals are gradually hydrolysed, associated with leaching of S. In the mature Australian soil, most of these minerals had already been transformed to goethite. Pipestems formed after roots in unripe soil are sites for synthesis and hydrolysis of minerals and serve as routes for atmospheric oxygen into the reduced subsoil and for soluble reaction products to exit the soil. Pipestems have an important role in the ripening of the soil profile.

KEYWORDS

jarosite, pedogenesis, scanning electron microscopy, schwertmannite, soil ripening, sulfidic material

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1 | INTRODUCTION

Sediments and aeolian deposits are often rather homogeneous parent materials to start with but, after exposure to soil forming processes, redistribution of elements takes place and increases heterogeneity. Reduced sediments can be exposed to atmospheric oxygen (O_2) through artificial drainage associated with agriculture or municipal development projects and dredging of waterways. These activities give rise to rapid changes in hydrology and result in drying of the soil, consequent irreversible loss of volume, cracking and formation of soil structure. Increased drainage can also take place, even though more slowly, by isostatic land uplift which still continues in Northern Europe after the Weichselian glaciation that ended about 10,000 years ago (Tikkanen & Oksanen, 2002).

Wetland plant species, such as reeds and rice, are capable of growing their roots into the reduced sediment (Khan et al., 2016; St-Cyr & Crowder, 1988) and emit O_2 radially into the rhizosphere (Maisch et al., 2020). Upon death and decomposition of most of the root tissue, large unoccupied pores persist down the soil profile. During times of lowered water tables, these macropores, due to their size, are the first soil pores to empty creating pathways for O_2 to diffuse very quickly down the soil profile. In this way, aerobic cylinders are created in the reduced matrix of the capillary fringe, which disturbs the redox state and results in oxidising reactions, which have particularly pronounced effects in acid sulfate (AS) soil materials. By contrast, the diffusion of O_2 through saturated AS gel materials is a very slow process (Ward et al., 2004).

In predominantly waterlogged soil, brown coatings on root surfaces, pore linings around former root channels (Chen et al., 1980; Richardson & Hole, 1979) and iron (Fe) concretions (Schroeder & Schwertmann, 1955; Yu et al., 2024) are common. They are formed after O_2 enters the root channel and gives rise to the oxidation of Fe^{2+} of the reduced soil to Fe^{3+} just around the root channel (Dent, 1986, p. 78), with subsequent precipitation of various ferric compounds, depending on the pH and Fe^{3+} activity of the immediate environment (Burton et al., 2007; Ward et al., 2004). Precipitated Fe compounds cement the soil matrix around the macropores upon drying, creating vertically aligned, cylindrical pedofeatures called 'pipe-stems'. Macropores, including root channels and desiccation cracks, are the most intensive sites for biogeochemical processes (Högfors-Rönholm et al., 2024).

Pipestems are prevalent in, but not limited to, AS soils where, upon oxidation of sulfidic material, sulfur (S) and Fe are released into the soil solution from pyrite (FeS_2) or iron monosulfidic compounds. Low pH in the oxidised AS horizons promotes weathering of soil minerals and dissolution of elements, which results in high electrical

Highlights

- Redistribution of elements upon ripening of sulfidic sediment material was studied.
- Jarosite and schwertmannite accumulate in pipestems around and in previous root channels.
- Upon hydrolysis and leaching, these minerals are converted to goethite.
- Pipestems have an important role in element fluxes and soil ripening.

conductivity in the soil solution and substantial leaching of elements (particularly Al, SO_4 -S) out of the solum (Österholm & Åström, 2004). In AS soils, conditions are favourable for the formation of new minerals such as schwertmannite, $Fe_8O_8(OH)_6(SO_4) \times nH_2O$, and jarosite, $KFe_3(OH)_6(SO_4)_2$, as metastable products of oxidation of sulfidic compounds. These minerals are temporary storages of residual acidity (Sullivan et al., 2012). Even when sulfidic materials are completely oxidised from a given horizon, jarosite and schwertmannite contain acidity that can be released at later stages of pedogenesis. When the oxidation front continues to extend to deeper horizons, for example, by artificial drainage and drought periods, the upper horizons are leached and these minerals are gradually converted to other minerals, such as ferrihydrite and goethite, simultaneously releasing the residual acidity.

Minerals mentioned above give the soil a distinctive pattern of contrasting colours: grey (or black if monosulfidic material is present) in the reduced matrix, and pale yellow and reddish brown in the pipestems and aggregate surfaces that represent the oxidised part of the soil (Yu et al., 2024). Morphological descriptions show that reduced and oxidised pockets coexist in horizons where sulfidic material is actively oxidising. The pale yellow colour of jarosite is commonly documented in AS soils (Dent & Pons, 1995), in both Finland (Yli-Halla, 1997) and Australia (Walker, 1960). Schwertmannite was first identified in sulfide-bearing mine sites in Finland (Bigham et al., 1994) and later extensively in mining sites around the world (Paikaray, 2021, with references) and it is a prevalent mineral in AS soil landscapes as well (Burton et al., 2007; Sullivan & Bush, 2004; Yu et al., 2024).

In studies on the composition of Fe plaques formed around the roots of wetland plants (Chen et al., 1980; Mendelsohn & Postek, 1982), lepidocrocite and goethite have been identified. However, detailed investigations of pipestems have not been undertaken to date. In this study, pipestems and the adjacent soil material from two

AS soils of different ages originating from highly different climates were characterised. The pipestem properties were documented at the microscopic scale at various stages of their life span from an unripe to a ripe soil. These observations contribute to our understanding of the ripening process and show how redox- and pH-active elements can be redistributed in AS soil as a response to vegetative growth, drainage and associated redox reactions.

2 | MATERIALS AND METHODS

2.1 | Description of the experimental sites

Two soils were sampled for this study, Ylistaro soil from Finland and Tweed Heads soil from Australia. The Ylistaro pedon (62° 55'N, 22° 29'E) in western Finland is on a flat coastal plain with an elevation of 26 m above the current sea level. Owing to the post-glacial land uplift, the soil surface emerged from the Baltic Sea at about 2000 years ago. Mean annual and summer air temperatures are 3.2 and 13.1°C, respectively. Average precipitation in 1991–2020 in a station 40 km away was 581 mm, and the average duration of snow cover was 143 days. The soil has a cryic temperature regime and according to the World Reference Base for Soil Resources (WRB) system (IUSS Working group WRB, 2022) it belongs to Thionic Gleysols (Siltic, Drainic, Humic, Bathysulfidic). This silty clay loam soil is artificially drained with subsurface tile installed to the depth of 1.0–1.2 m. It has an agricultural history of about 150 years for growing small grains (oats, barley), grasses and potato. Soil properties determined in earlier studies are shown in Tables S1 and S2.

The Tweed Heads soil in the McLeod's Creek (28°18'S, 153°30'E) has an elevation of 0.5 m above the current sea level and lies in an AS landscape within the Tweed River catchment in north-eastern New South Wales, Australia. For soil properties, see Tables S1 and S3. The Tweed River floodplain is a drowned river valley that formed post-sea level rise following the landward movement of marine sands from offshore and clayey sediments into the prior valleys and the formation of sand barriers (Roy et al., 1980). The basin behind the sand barriers progressively infilled with muddy estuarine material to form floodplains characterised by low elevation (<1 m above the current sea level) backswamps grading up into well-developed levees formed by fluvial deposition. The soils forming on these floodplains characteristically contain appreciable Fe sulfide minerals. The Tweed Heads site has a subtropical climate with an average annual precipitation of 1572 mm, an average annual maximum and minimum

temperature of 25.8 and 14.5°C, respectively, at nearby Murwillumbah. During low rainfall periods, water table depth is controlled by evapotranspiration, which lowers the water table sufficiently to expose the sulfidic subsoil layers to oxidation (Wilson et al., 1999). Additionally, anthropogenic disturbances through drainage, peat burning and ploughing have also caused the oxidation of the sulfidic subsoils (Macdonald et al., 2004). These soils are classified as extratidal sulfuric hydrosols (Isbell, 2023).

2.2 | Sampling and methodology

The Ylistaro pedon was sampled for pipestems (Figures 1a and S1a) and the adjacent soil material from 60 to 215 cm. There were no pipestems in the upper soil layers (0–60 cm) and the reduced sulfidic subsoil below 200 cm (Figure S1b). Soil samples from different depths were taken with the Ejkelkamp Edelman combination auger. Pipestems from the interiors of large prismatic aggregates were picked, and the adjacent grey soil material was sampled with a knife. The pipestems were allowed to air dry while the soil matrix samples were freeze-dried to prevent the oxidation of the sulfidic material. The pH (1:1) of 170–215 cm fresh matrix samples, kept at 5°C, was measured 2 days after sampling and again after 8 weeks of aerobic incubation. The Ylistaro samples were gamma-irradiated upon shipment to Australia for analysis.

The Tweed Heads sampling site was next to a drainage channel in a sugar cane field. The water level in the channel was at a depth of 120 cm, and the upper drain sediment was monosulfidic black ooze (Sullivan et al., 2018). After opening a fresh soil profile on the ditch slope, the pH was measured in situ at 10 cm intervals and soil was sampled for pipestems (Figure 1b) and the adjacent soil material at 35–55, 55–75, 75–95 and 95–130 cm, which was the lowest depth where pipestems were found. The upper three soil layers were oxidised, and the lowest one was partially oxidised, with no mottling in the soil matrix. The samples were packed airtight in plastic bags. The pipestems were allowed to air dry in room temperature, and the adjacent soil material was dried at 40°C.

For the chemical analyses, the samples were ground with an agate mortar and pestle. Before grinding, the bulk soil attached to the pipestems was gently removed in the laboratory using a small knife. The bulk soil samples were extracted with deionised water (1:5, w:v) to get an estimate of water soluble ions (Ca, Mg, Na, K, Al, sulphate and chloride) in the respective soil layer. The total C and S concentrations of all samples were measured by dry combustion (LECO 220 sulfur/carbon analyser). Poorly crystalline Fe and Al (hydr)oxides were extracted

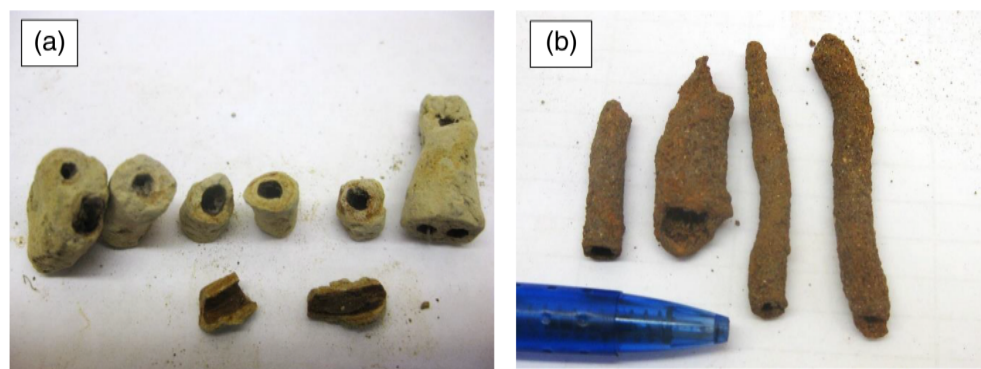


FIGURE 1 Selected pipestems (a) from 170 to 190 cm depth layer of the Ylistaro soil and (b) from 80 to 90 cm depth layer of the Tweed Heads soil. Photos by Markku Yli-Halla.

with 0.2 M ammonium oxalate solution, pH 3.0, for 2 h in the dark. Crystalline Fe oxides were dissolved with citrate–dithionite–bicarbonate treatment in a water bath at 80°C. Fe and Al were measured with Inductively Coupled Plasma Optical Emission spectroscopy (ICP-OES). The elemental concentration data were tested with one-way analysis of variance and Tukey's honestly significant difference (HSD) test at $p = 0.05$.

The micromorphology of the pipestem samples was investigated with a scanning electron microscope (SEM) (Stereoscan 440, Leica, Cambridge Ltd, UK). Samples were mounted on an aluminium stub and coated with carbon. SEM images were taken using a SEM (LEICA, Stereo Scan 440, working distance [WD] = 7 mm, probe 40 pA, electron high tension [ETH] = 10 kV). The elemental/atomic composition of selected areas of the samples was determined using an ISIS energy dispersive x-ray EDX0 micro-analysis system (Oxford EDS detector, PBQuant Software, WD 25 mm, Probe 500 pA, ETH = 20 kV) using a peak-to-background method (Sullivan & Bush, 1997).

The mineralogy of the samples was investigated with powder x-ray diffraction (XRD). For the analysis, the samples were hand ground ($<63 \mu\text{m}$) using an agate mortar and pestle and micronised using propanol as the solvent agent and then suspended in ethanol which was allowed to evaporate. X-ray diffractograms were obtained for randomly oriented powders using a Bruker D4 diffractometer using $\text{Co K}\alpha$ radiation (Bruker AXS GmbH, Germany). The diffraction patterns were recorded from 5 to $80^\circ 2\theta$ in steps of $0.05^\circ 2\theta$ step size with 2 s counting time per step. The XRD patterns were evaluated using EVA software (DIFFRAC-plus evaluation package, Bruker AXS, Germany).

3 | RESULTS

3.1 | Soil pH

The upper layers of the Ylistaro soil, down to 190 cm, were very acidic (Figure 2a). The lowest pH of 3.4 at 100–

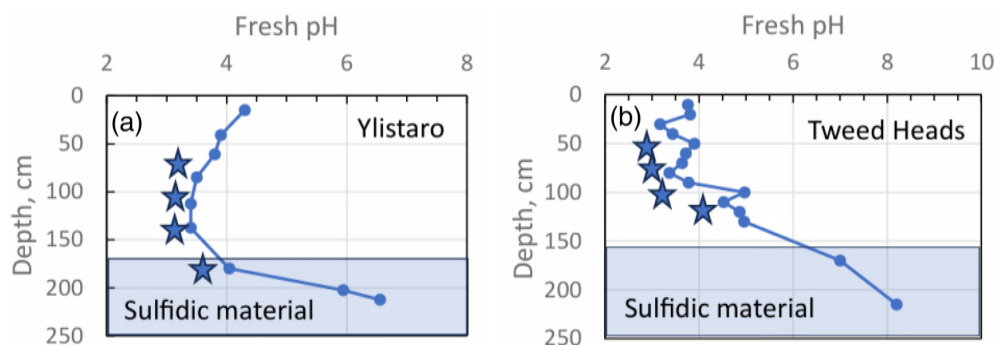
150 cm suggests that sulfidic material had recently been oxidised. Earlier studies of soil layers down to 150 cm (Yli-Halla, 1997) have shown that there was no decrease in pH upon aerobic incubation, indicating that no sulfidic material remained in these layers. However, in spite of a low pH (4.0), sulfidic material remained at 170–190 cm, and definitively in deeper layers, where the pH reached neutrality (Figure 2a, Table S1). The Tweed Heads soil was very acidic with the minimum in situ pH of 3.17 at 30 cm (Figure 2b). The pH increased to values close to 5 in the grey soil below 100 cm. Although none of the Tweed Heads soil layers sampled for this study contained sulfidic soil material, the soil profile data typical of this site (Table S1) indicate that a sulfidic subsoil layer immediately underlies the sampled soil layers.

3.2 | Morphological characteristics of the pipestems

The Ylistaro pipestems (Figure 1a) were thinnest (5.6–6.0 mm) at 60–80 cm while the thickness increased systematically downwards to the maximum of 7.3–8.0 mm at 170–190 cm. All pipestems had a hole of 1–3 mm in diameter, getting narrower at greater depths. The walls of the pipestems were thinnest (1.1–1.7 mm) at 60–110 cm and substantially thicker (3.1–3.8 mm) further down, particularly at 170–190 cm. In the Tweed Heads soil (Figure 1b), the pipestems were narrower (4.3–5.8 mm) than in Ylistaro with no distinct difference in thickness between the depths; the central holes had diameters of 2.1–2.9 mm, and the walls (1.1–1.5 mm) were thinner than in the Ylistaro pipestems. In spite of this difference, the Tweed Heads pipestems were noticeably stronger and could be sampled as longer pieces than those from Ylistaro.

The pipestem matrix in the Ylistaro soil had a brownish (10YR 5/3) colour while the adjacent soil material was grey (5Y 5/1). In the pipestem central holes at 60–150 cm, there was a dark reddish brown (5YR 3/3)

FIGURE 2 Fresh pH of the Ylistaro soil (a) and the Tweed Heads soil (b). The * sign indicates the depth of the pipestems that were studied.



coating completely covering the surfaces of the hole. The Tweed Heads pipestem matrices were dusky red or dark reddish brown (2.5–5YR 3/4). A darker coating on the inner hole surfaces was distinguished also in these pipestems.

3.3 | Mineralogical and elemental concentrations

According to the XRD analysis, the Ylistaro pipestems and the adjacent soil material largely consisted of quartz and weatherable minerals such as different feldspars, phlogopite, chlorite and hornblende. The pipestems at 60–150 cm, but not at 170–190 cm, contained jarosite while pyrite was detected in the soil matrix at 170–215 cm. In the Tweed Heads soil, quartz, plagioclase, kaolinite and goethite were detected but easily weatherable minerals were absent. The pipestems between 30 and 95 cm, but not 95–130 cm, contained traces of jarosite, while no pyrite was detected at any investigated depth in this soil.

In the oxidised horizons of Ylistaro, that is, down to 150 cm, S concentration of the pipestems (Table 1) was 8–12 times higher than in the adjacent soil material. In the reduced soil below 170 cm, total S concentration was much higher, and 67% of it was still in the sulfidic (Cr reducible) form. In the largely reduced 170–190 cm layer, the pipestems had a slightly lower S concentration than the adjacent soil material, where the S concentration was clearly higher than in the upper layers. The lowest S concentration of the pipestems in the Ylistaro soil thus occurred in this horizon richest in S among the horizons that contained pipestems. In the Tweed Heads soil, the pipestems mostly contained less S than the Ylistaro pipestems whereas in the adjacent soil material, the S concentrations were similar to those in the oxidised horizons of Ylistaro. It is noteworthy that the S concentration of the pipestems systematically increased upwards in both soils and so did the degree to which S had enriched in the pipestems compared to the adjacent soil material.

Enrichment of S in the pipestems of the Tweed Heads soil was less pronounced than in the Ylistaro soil (Table 1).

Among the elements investigated, Fe was by far the most enriched in the pipestems in both soils. In the Ylistaro and Tweed Heads soil, the enrichment of Fe_{cdb} was 14- to 92-fold and 13- to 27-fold, respectively, as compared to the adjacent soil material (Table 1). Both the pipestems and the adjacent soil material of the Tweed Heads soil contained more Fe_{cdb} than the Ylistaro samples. In the Ylistaro pipestems, as much as 86%–90% of Fe_{cdb} was extractable with the oxalate solution while the corresponding percentage was only 15%–34% in the Tweed Heads pipestems. In Ylistaro, the extractability of Fe_{cdb} increased from 62% to 100% with depth while the opposite (from 74% to 38%) was observed in Tweed Heads. On the molar basis, the pipestem Fe_{ox} concentration was 9.5-fold compared to Al_{ox} while the concentration of Fe_{ox} in the adjacent soil material was only 1.7-fold compared to Al_{ox} . Enrichment of Al in the pipestems was thus much less pronounced than that of Fe. The range of C concentrations in all the pipestems was rather narrow, 0.9%–1.9%. The pipestems usually contained more C, up to 2.4-fold, than the respective adjacent soil material (Table 2) presumably due to the presence of root remnant materials.

The data allow to make estimations of the degree of S depletion that has taken place over time. According to Österholm and Åström (2002), it was assumed that the soil at 50–150 cm had originally contained the same S concentration as the reduced parent material at 210–215 cm (Table 1). The bulk density is assumed to be 0.9 kg dm^{-3} (Österholm & Åström, 2002). This calculation suggests a 70 ton ha^{-1} removal of S from the soil matrix at 50–150 cm, part of which remains in the coatings and pipestems. If the ratio of FeS and FeS_2 in the parent sediment material was 1:1, about 80 ton ha^{-1} Fe has been mobilised from the soil matrix. In the Tweed Heads soil, a comparable calculation could not be done because the original sulfide composition of the reduced parent material was not available.

TABLE 1 Selected chemical characteristics of the pipestem samples (P), adjacent soil material (M) and enrichment factors (P/M). The chromium reducible (Cr-S) values are for the adjacent soil material. Y=Ylistaro, T = Tweed Heads, cdb = citrate-dithionite-bicarbonate extraction, ox = oxalate extraction.

Depth, cm	C, %			S, g kg ⁻¹			Fe-cdb, g kg ⁻¹			Fe-ox, g kg ⁻¹			Al-ox, g kg ⁻¹			Cr-S g kg ⁻¹
	P	M	P/M	P	M	P/M	P	M	P/M	P	M	P/M	P	M	P/M	
Y60–80	1.6 ^c	0.8 ^b	2.0	11.5 ^g	0.9 ^a	12.8	68.5 ^b	2.9 ^b	23.6	60.3 ^f	1.8 ^b	33.5	2.7 ^e	0.6 ^a	4.5	n.d.
Y90–110	1.4 ^{bc}	1.0 ^c	1.4	10.1 ^f	1.3 ^a	7.8	55.8 ^{ab}	4.0 ^b	14.0	50.1 ^e	3.2 ^c	15.7	2.4 ^d	0.7 ^{ab}	3.4	n.d.
Y130–150	1.4 ^{bc}	1.2 ^d	1.2	8.5 ^e	1.2 ^a	7.1	45.6 ^a	2.9 ^b	15.7	39.2 ^{cd}	2.4 ^b	16.3	2.3 ^d	0.7 ^{ab}	3.3	0.04 ^a
Y170–190	1.5 ^{bc}	1.5 ^e	1.0	5.2 ^c	7.8 ^b	0.7	46.2 ^a	0.5 ^a	92.4	41.5 ^d	0.5 ^a	83.0	2.3 ^d	1.0 ^c	2.3	5.23 ^b
Y210–215	-	1.6 ^e	-	-	9.1 ^c	-	-	2.1 ^b	-	-	2.2 ^b	-	-	0.6 ^a	-	6.17 ^c
T30–55	1.9 ^d	0.8 ^b	2.4	7.8 ^d	1.0 ^a	7.8	95.8 ^c	6.2 ^c	15.5	28.9 ^b	4.6 ^d	6.3	1.8 ^c	1.0 ^c	1.8	n.d.
T55–75	1.4 ^{bc}	0.7 ^b	2.0	4.8 ^c	0.8 ^a	6.0	99.2 ^{cd}	6.4 ^c	15.5	34.0 ^{bc}	4.2 ^d	8.9	1.9 ^c	1.0 ^c	1.9	n.d.
T75–95	1.2 ^b	0.8 ^b	1.5	3.2 ^b	0.8 ^a	4.0	112.1 ^{cd}	8.6 ^d	13.0	21.4 ^a	4.2 ^d	5.1	1.3 ^b	0.8 ^b	1.6	0.05 ^a
T95–130	0.9 ^a	0.6 ^a	1.5	1.1 ^a	0.6 ^a	1.8	114.9 ^d	4.2 ^b	27.4	17.7 ^a	1.6 ^b	11.1	0.7 ^a	0.6 ^a	0.9	0.04 ^a

Note: The mean values within a column marked with the same letter are not statistically different at $p = 0.05$.

TABLE 2 Elemental composition (%) of the pipestem matrices, interior coatings of the pipestem hole and the adjacent soil material on the basis of SEM–EDX analysis. Y=Ylistaro, T = Tweed Heads.

Depth (cm)	Pipestem matrix (adjacent soil material)				Pipestem interior coating			
	Fe	S	Si	Al	Fe	S	Si	Al
Y60–80	13.8 ^b (2.8)	2.1 ^d (0.2)	19.9 ^b (25.3)	4.9 ^{ab} (6.1)	42.7 ^b	4.2 ^b	1.3 ^a	1.1 ^{ab}
Y90–110	10.0 ^{ab} (3.7)	1.2 ^c	24.1 ^c (26.2)	5.7 ^{bc} (6.8)	53.0 ^b	4.8 ^b	1.5 ^a	1.9 ^{ab}
Y130–150	9.1 ^{ab}	0.81 ^{abc}	22.9 ^c	5.9 ^{bc}	43.7 ^b	5.3 ^b	3.9 ^a	1.5 ^{ab}
Y170–190	6.8 ^{ab}	0.50 ^{ab}	26.1 ^c	6.4 ^c	7.6 ^a	5.5 ^b	14.1 ^b	5.1 ^c
T30–55	39.9 ^d	0.91 ^{bc}	10.1 ^a	3.8 ^a	61.1 ^b	1.25 ^a	2.3 ^a	1.0 ^a
T55–75	30.8 ^c	0.64 ^{ab}	9.9 ^a	4.4 ^a	49.6 ^b	1.36 ^a	2.4 ^a	1.7 ^{ab}
T75–95	32.1 ^c	0.39 ^{ab}	12.2 ^a	4.6 ^a	54.5 ^b	0.60 ^a	2.3 ^a	2.1 ^b
T95–130	29.3 ^c	0.18 ^a	11.0 ^a	4.3	42.9 ^b	0.20 ^a	2.3 ^a	1.4 ^{ab}

Note: The mean values within a column marked with the same letter are not statistically different at $p = 0.05$. Data of the soil matrix were excluded from the test.

3.4 | Micromorphology and composition of the pipestems

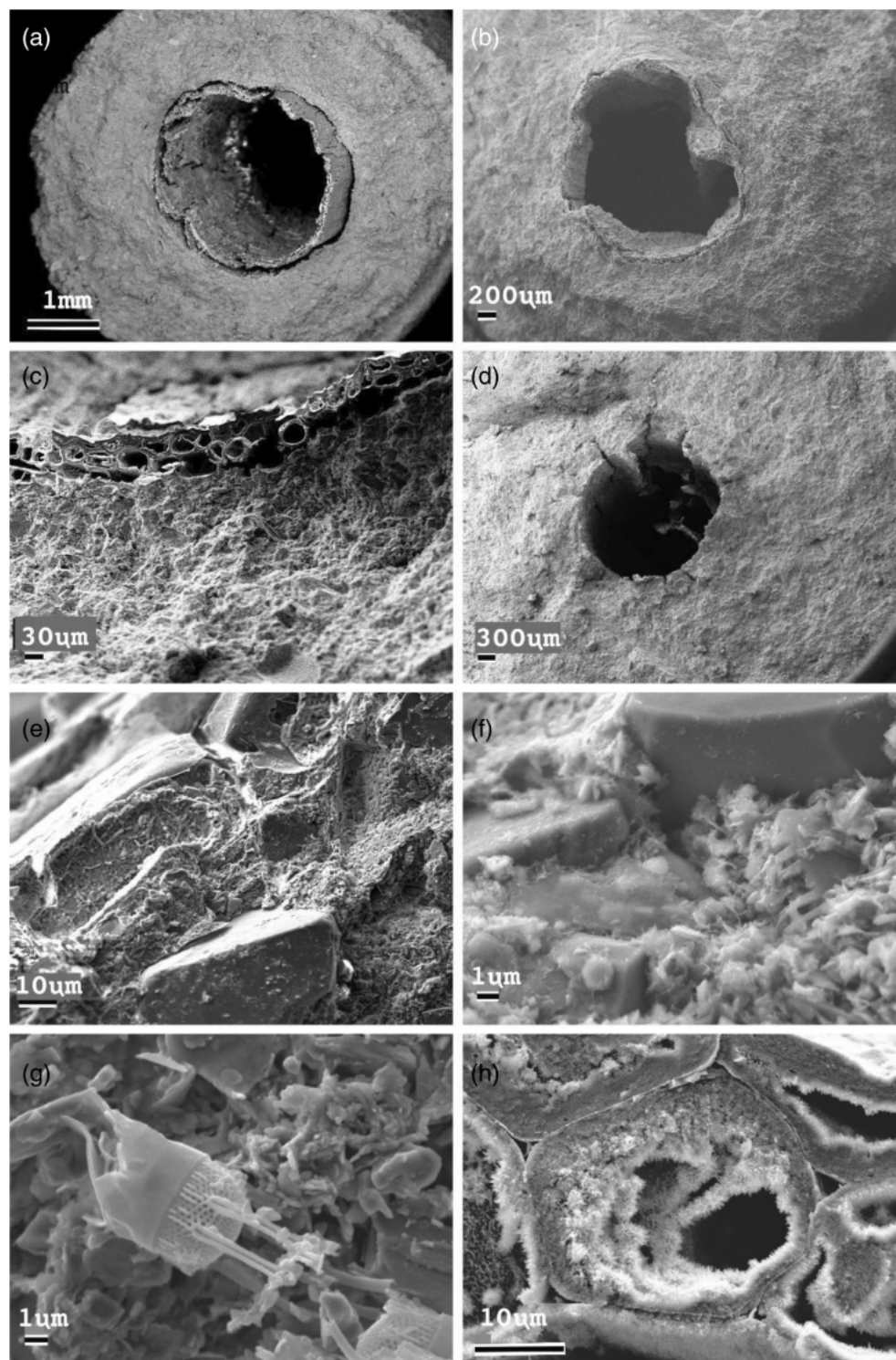
The pipestem matrices in cross-section from both soils had a dense compact appearance in the SEM (Figure 3a,b) often surrounding a hollowed-out former plant root with only from one to five layers of the outermost plant root cells ($10 \times 50 \mu\text{m}$) remaining intact (Figure 3c). The inner hole of some pipestems did not contain visible plant root remnants (Figure 3d).

The matrices of the Ylistaro pipestems had only slightly lower concentrations of Si and Al compared to the adjacent soil material (Table 2), and that was the case also with the other major constituents of silicate clays (K, Ca and Mg; Table S4). Molar ratio Si/Al (3.7–4.1) in

the pipestem matrices was the same as in the adjacent soil material at 60–110 cm, suggesting the similarity of these samples in terms of silicate mineral compositions. However, the Fe concentrations in the pipestem matrices were about three to four times greater than in the adjacent soil material. The pipestem matrices in the upper soil layers (60–150 cm) of the Ylistaro soil contained more Fe than the ones from the lowest horizon (170–190 cm; Table 2) by a magnitude of two at the maximum.

The total Fe concentration of the Tweed Heads pipestems was 3- to 6-fold compared to the pipestems at 60–150 cm of the Ylistaro soil. SEM images (Figure 3e, f) and the SEM–EDX data (Table 2) show that the pipestem bodies contained mineral grains similar to those of the adjacent soil materials cemented together by Fe

FIGURE 3 Scanning electron micrographs of (a) section across a pipestem from 60 to 80 cm depth layer of the Ylistaro soil showing an macropore coating composed of plant root cell remnants, (b) section across a pipestem from 170 to 190 cm depth layer of the Ylistaro soil showing an macropore coating composed of plant root cell remnants, (c) detail of the macropore coating composed of plant root cell remnants shown in (b), (d) section across an uncoated pipestem from 170 to 190 cm depth layer of the Ylistaro soil, (e) section across an impregnated plant cell adjacent the pipestem macropore (top left) and the surrounding soil matrix from 60 to 80 cm depth layer of the Ylistaro soil, (f) detail of a section across the dense impregnated soil matrix within the pipestem from 60 to 80 cm depth layer of the Ylistaro soil, (g) remnant of a diatom (*Skeletonema* sp.) in a section across the matrix of a pipestem from 170 to 190 cm depth layer of the Ylistaro soil and (h) plant root cell remnants adjacent the pipestem macropore with the cell walls coated with schwertmannite-like precipitates from 170 to 190 cm depth layer of the Ylistaro soil.



compounds. Frequent diatoms were observed in various soil layers in Ylistaro (Figure 3g) presumably originating from the soil parent materials.

The interior coatings of the central holes were morphologically most varied and spectacular. In the Ylistaro soil, these coatings consisted of well-preserved cell compartments that were completely or partly filled (Figure 3h). These infillings had a high concentration of

Fe (43%–53%) and S (4%–5%) and low concentrations of Al (1.1%–1.9%) and Si (1.3%–3.9%; Table 2) indicating all but the absence of silicate minerals. The infillings consisted of small ($\sim 2 \mu\text{m}$ in diameter) ‘whiskery’ balls similar in submicromorphology (Figure 4a–c) to that for schwertmannite accumulations from AS soil landscapes (Song et al., 2018; Sullivan & Bush, 2004; Vithana et al., 2015). The Fe and S concentrations of the coatings

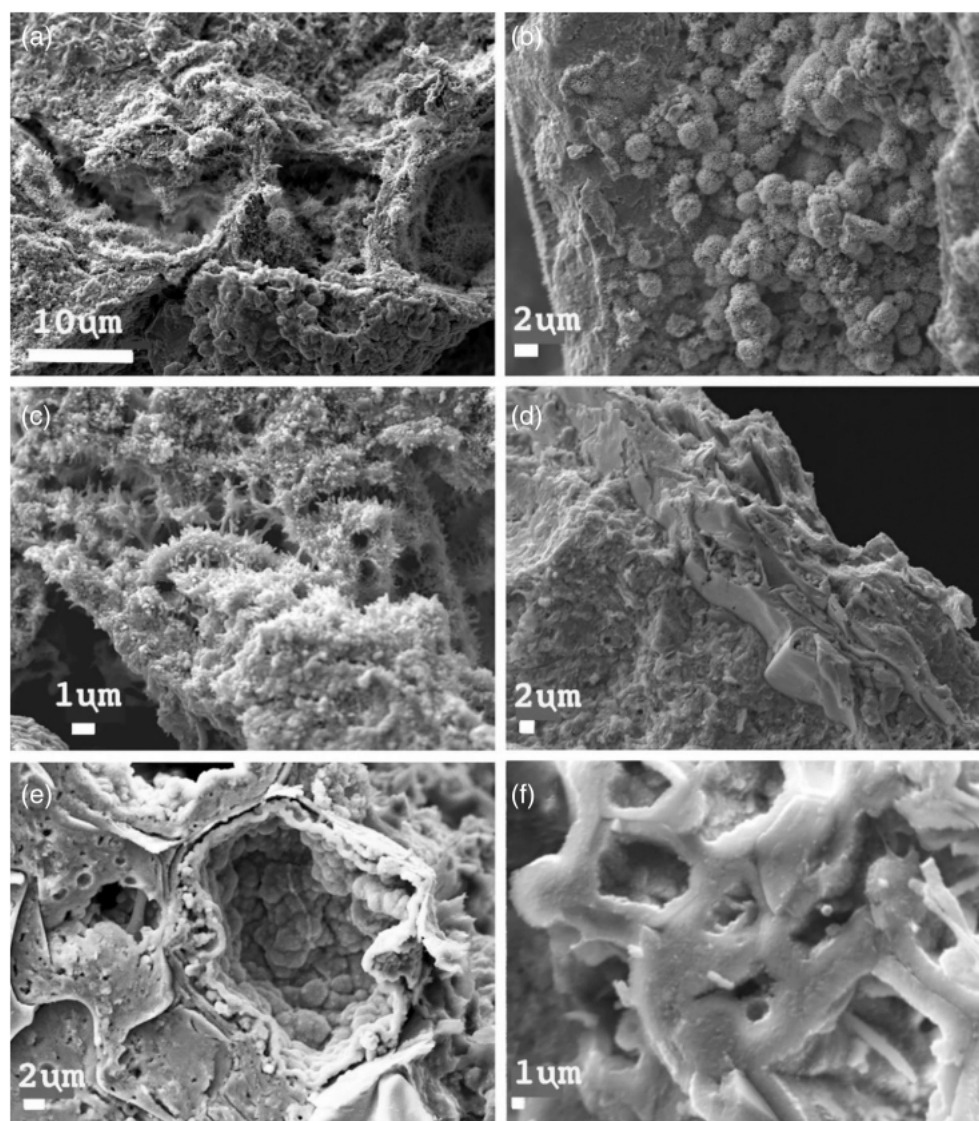


FIGURE 4 Scanning electron micrographs of (a–c) plant root cell remnants adjacent to the pipestem macropore with the cell walls coated with schwertmannite-like precipitates from 130 to 150 cm depth layer of the Ylistaro soil, (d) plant root cell remnant adjacent the pipestem macropore (upper right) with the cell walls completely filled with Fe-rich precipitates from 75 to 95 cm depth layer of the Tweed Heads soil, (e) plant root cell remnants adjacent the pipestem macropore with the cell walls coated with Fe-rich precipitates from 75 to 95 cm depth layer of the Tweed Heads soil and (f) plant root cell remnants adjacent the pipestem macropore with the cell walls coated with iron-rich precipitates from 55 to 75 cm depth layer of the Tweed Heads soil.

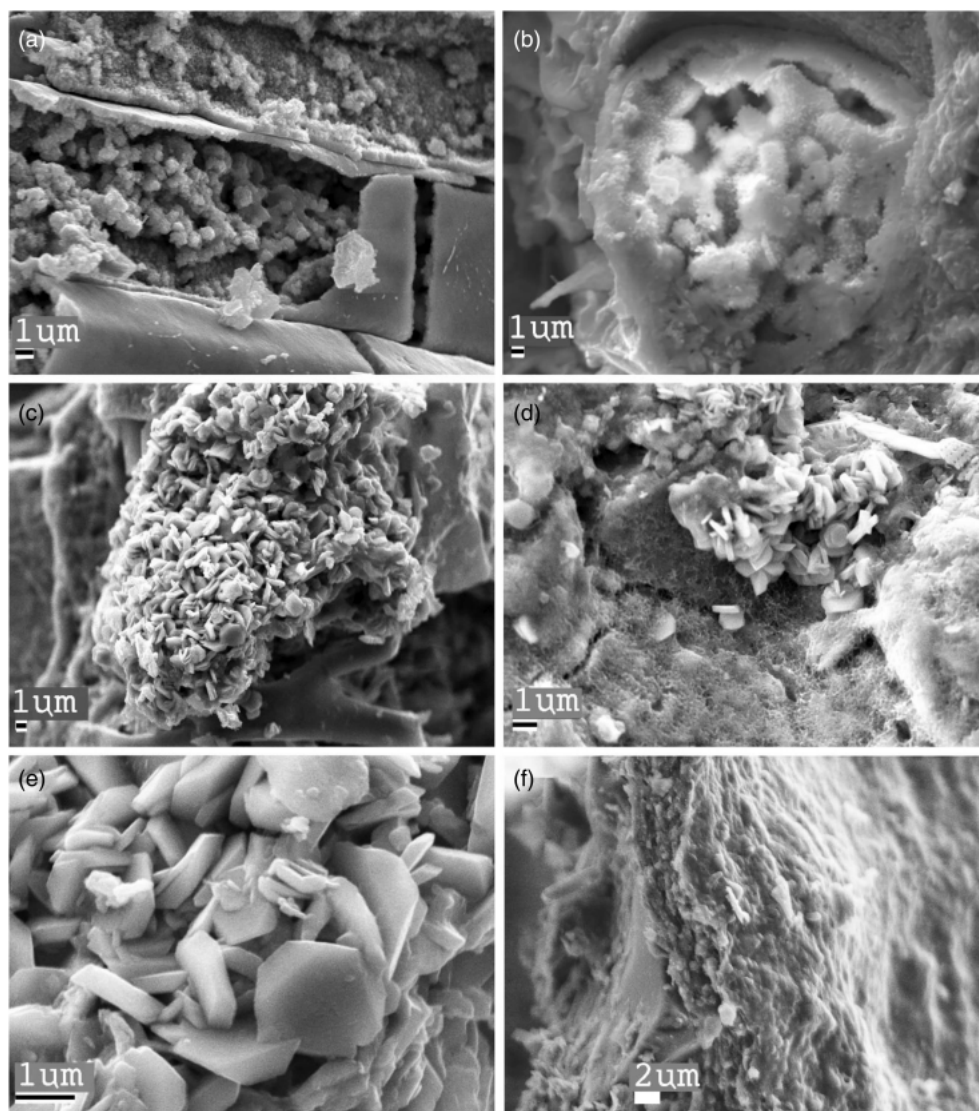
in the Ylistaro soil also resembled those of schwertmannite (Fe 58%, S 4.2%) but the molecular ratios Fe/S of 4.8–6.5 were intermediate of those in schwertmannite (ideally Fe/S 8) and jarosite (1.5).

Also in the Tweed Heads pipestems, there were plant cell-like features in the pipestem holes but they were commonly fragmented and partly collapsed (Figure 5a,b). Many pipestem holes contained occasional mineral grains, and illuvial soil material had accumulated on the walls of the central hole coating the collapsed concentric ring of cells (Figure 4d). In Tweed Heads, the interior coatings around the pipestem central hole (Figures 4e and 5b) were richer in Fe than the pipestem matrices, corresponding to 68%–97% of goethite and not leaving much room for silicate minerals. In these infillings, there was a trend of decreasing Fe concentration with depth. Sulfur concentration of the coatings/infillings in the Tweed Heads soil was higher than in the pipestem matrices (except the lowest

soil layer 95–130 cm) but clearly lower than in the Ylistaro soil. The Fe precipitate accumulations in the root remnants lining the interior hole of the Tweed Heads pipestems had similar micromorphology to those observed in the Ylistaro soil (Figure 3h). Although the whiskery ball submicromorphology for schwertmannite was also frequently observed in the SEM images of the Tweed Heads pipestem coatings (Figure 5a,b), these may be goethite pseudomorphs after schwertmannite given the strong and commonly observed tendency for schwertmannite to convert to goethite (Sullivan & Bush, 2004): this may well explain the relatively low S contents of the coatings within the Tweed Heads pipestems. Jarosite crystals were only infrequently observed in the pipestems of the Ylistaro soil at 60–150 cm soil (Figure 5c–e) and not observed at all in those of the Tweed Heads.

Differing from the pipestems above, the dark brown coatings of the pipestem hole at 170–190 cm in the

FIGURE 5 Scanning electron micrographs of (a) plant root cell remnant adjacent to the pipestem macropore with the cells partially infilled with schwertmannite-like precipitates from 30 to 55 cm depth layer of the Tweed Heads soil, (b) plant root cell remnant adjacent to the pipestem macropore with the cells partially infilled with schwertmannite-like precipitates from 55 to 75 cm depth layer of the Tweed Heads soil, (c) segregation of jarosite crystals on a pipestem macropore from 130 to 150 cm depth layer of the Ylistaro soil, (d) segregation of jarosite crystals (upper right) on a pipestem macropore from 80 to 90 cm depth layer of the Ylistaro soil, (e) detail of (c) showing euhedral jarosite crystals and (f) thin illuvial coating of aluminosilicate clay (centre) on the surface of a pipestem macropore from 170 to 190 cm depth layer of the Ylistaro soil.



Ylistaro soil were thinner and were not always composed of plant cell-like structures and contrasted in composition with the thicker Fe-enriched coatings of the central hole in the pipestems located in the upper soil layers (Figure 5f). These coatings were much lower in Fe and higher in Si and Al (Table 2), suggesting the presence of illuvial clay.

3.5 | Composition of water extracts

Among the cations, Ca and Mg dominated in the water extracts of Ylistaro while Na was by far the most abundant in the extracts of the Tweed Heads soil (Table 3), reflecting the dominant influence of marine-derived salt in this coastal landscape. In the most acidic horizons (pH 3.4–3.5) of Ylistaro, there was considerable Al in the water extracts but in spite of equally low pH in the Tweed Heads soil, dissolved Al was nearly absent there.

The total amounts of cations and anions matched well in both soils. In Ylistaro, sulfate was the dominant anion while chloride was below the detection level (50 mg l^{-1}) while in the Tweed Heads soil, chloride was abundant in the lower horizons. In order to extrapolate from the soil-to-water ratio of 1:5 to that of 1:0.5, attempting to mimic pore water concentrations, the results were simply multiplied by 10. This calculation suggests that in the upper horizons examined, the concentration of ions was at the usual level, which is measured in soil solution (0.02 M), while in the lower horizons, the concentrations were between 5- and 10-fold. An estimation of sulfate concentration in pore water shows that in Ylistaro, the concentration was $2500\text{--}4000 \text{ mg l}^{-1}$ in the horizons which contained sulfidic material but was much lower in the oxidised horizons above. In the Tweed Heads soil, the concentration of sulfate was almost similar in the extracts of all soil layers.

TABLE 3 Sum of cations and anions and the proportion of cations in the water extracts of the soil matrix samples from Ylistaro (Y) and Tweed Heads (T) at soil-to-water ratio of 1:5 and extrapolated estimates for total ion and sulfate concentrations at soil-to-water ratio of 1:0.5.

Depth (cm)	Cations at 1:5 cmol _c l ⁻¹	Anions at 1:5 cmol _c l ⁻¹	Estimate of ions at 1:0.5 cmol _c l ⁻¹	SO ₄ estimate at 1:0.5 mg l ⁻¹	Ca %	Mg %	K %	Na %	Al %
Y60–80	0.64 ^a	0.85 ^a	0.015	410 ^a	40	31	12	9	8
Y90–110	1.20 ^{ab}	1.27 ^{ab}	0.025	608 ^{ab}	36	30	11	11	12
Y130–150	2.32 ^b	2.44 ^b	0.048	1170 ^c	30	37	6	15	13
Y170–190	5.49 ^c	5.14 ^c	0.106	2468 ^d	25	44	6	23	3
Y210–215	8.70 ^d	8.33 ^d	0.170	3999 ^e	23	42	6	29	0
T30–55	2.20 ^b	1.37 ^{ab}	0.042	657 ^{abc}	3	9	1	86	1
T55–75	2.46 ^b	1.01 ^{ab}	0.041	485 ^a	2	7	2	88	1
T75–95	6.70 ^c	7.31 ^d	0.135	1055 ^{bc}	1	5	2	92	0
T95–130	11.40 ^e	9.51 ^e	0.209	885 ^{abc}	1	5	2	90	2

Note: The mean values within a column (cations, anions and SO₄) marked with the same letter are not statistically different at $p = 0.05$.

4 | DISCUSSION

Pipestems were not found in the top layers that are affected by most biological activity, shrink-swell disturbance and agricultural operations in both sampling sites and Ylistaro also by the annual frosturbation. They occurred in the ripe subsoil horizons, and in Ylistaro also in the BC horizon still consisting of sulfidic material. Oxidised zones have gradually extended radially around the root holes, resulting in the oxidation of Fe²⁺ to Fe³⁺ and the cementation of soil material to form the pipestem. From visual observations (see Figure S1), it is obvious that in the unripe sediment, pipestems are formed first, and desiccation cracks appear only later after the plants have extracted water from the unripe horizon. We assume that the oldest pipestems are closest to the soil surface while the deepest ones are the youngest developing alongside the progressing drainage and the lowering oxidation front. Most of the pipestems may have been formed in root channels created by reedy vegetation (most likely *Phragmites australis*) or other plants capable of penetrating the reduced sediment. The pipestems are thus assumed to have formed before the agricultural use of these soils.

In the Ylistaro soil, the pipestems at 170–190 cm are at an early stage of formation, surrounded by sulfidic soil. These pipestems had a much lower Fe concentration than the pipestems above, which indicates that (1) oxidation of sulfidic material has not yet generated much Fe to be mobilised and precipitated forming pipestems, or (2) the conditions well below the drainage pipes have been too reduced to allow appreciable oxidation of Fe sulfides and subsequent mobilisation and precipitation of Fe³⁺ compounds.

Some conclusions can be made about the origin of Fe in the pipestem. The concentration of Cr reducible S in Ylistaro at 170–190 cm was 5.23 g kg⁻¹ (Table 2), and assuming that the ratio of FeS/FeS₂ is 1:1, the bulk soil material contains 6.0 g kg⁻¹ (0.6%) Fe. In the pipestems of this soil layer, the Fe concentration was 6.8%, that is, about 11 times more than would be the case if all Fe originated from the oxidising sulfidic material within the pipestem matrix itself. This simple calculation suggests a flux of Fe from the surrounding soil already at an early stage of pipestem formation. This proposition is strongly supported by observations of the oxidation process of AS materials in the laboratory under controlled conditions (Ward et al., 2004) indicating that the diffusion of O₂ into reduced sulfidic soil material adjacent to a macropore will initiate sulfide oxidation. Whilst SO₄-S derived from the oxidation of pyrite from AS materials is soluble, the solubility of the Fe released from such oxidation is appreciable only at pHs < ~4.0. Given the very acidic conditions in almost all soil layers examined in this study, both the SO₄-S and Fe derived from the sulfide oxidation would be able to relocate from the zones of oxidation towards the root holes where the formation of minerals containing SO₄-S and Fe (e.g., schwertmannite and jarosite) is favoured.

The relative accumulation of S in the pipestem was much less pronounced than that of Fe. Assuming that in the original unripe state, the 60–80 cm layer had the same concentration of Cr reducible S as the currently unripe 210–215 cm layer (0.62%), and the composition of the pipestems at 60–80 (1.15% S) suggests a 1.9-fold accumulation of S. While in the Ylistaro parent sediment at 210–215 cm, the mass ratio of Fe:S was only about 1.1

(assuming FeS/FeS₂ ratio of 1:1), the Fe:S mass ratio of 6 in the Ylistaro pipestems at 60–80 cm suggests accumulation of Fe in the pipestems and leaching of SO₄²⁻ out of the soil profile, supported also by the low soluble S concentrations in the oxidised horizons (Table S3) and substantial S loading and minor Fe loading from AS soils (Österholm & Åström, 2004). Part of the S remaining in the oxidised horizon has been incorporated in new minerals such as jarosite and schwertmannite, explaining the relatively high S concentration in the pipestems of 60–110 cm as compared to the S-depleted adjacent soil material. At 170–190 cm on the contrary, the pipestems were lower in S than the adjacent soil material, due to the abundance of remaining sulfidic material.

In addition to the impregnations forming the pipestem matrix of the Ylistaro soil, a new solid phase consisting of Fe-S compounds seems to have formed around the pipestem hole within the remaining cell walls of the cortex. Low Si and Al concentrations in these coatings suggest that they have been formed from elements released from the surrounding soil into the pipestem hole. Similar features in root channels were reported by Mendelssohn and Postek (1982) who proposed that lepidocrocite and goethite had been precipitated within the outer epidermal cells as a consequence of radial O₂ loss from the roots. This mechanism of formation would require that the initiation of the present pipestems of these interior coatings occurred prior to the current (agricultural) vegetation as the roots of these vegetation types do not grow roots to these great depths.

The Tweed Heads site represented a more mature soil indicated by the lack of weatherable minerals and the high crystallinity of the Fe oxides, as shown by the ratio of oxalate and citrate–dithionite extractable Fe. High concentration of Fe in the pipestem matrices indicates that large amounts of Fe have been mobilised and translocated within the soil profile. Equal Fe concentration in the Tweed Heads pipestems of all horizons suggests that these horizons were at a similar state of pedogenic development. Very low soil pH shows that this soil has also contained sulfidic material that has oxidised completely, with no Cr reducible S and little water-extractable SO₄-S remaining in the soil material adjacent to the pipestems. The pipestems contained only traces of jarosite, which indicates that the metastable oxidation products of FeS₂ (including schwertmannite) had already been hydrolysed and converted to goethite. Even though the pipestem S concentration in Tweed Heads soil was in most cases lower than in Ylistaro, the pipestems in both soils contained much more S than the oxidised soil around the pipestems (excepting 170–190 cm in Ylistaro). The soil matrix is thus more rapidly depleted of S than that of the pipestems, owing to the metastable S-containing minerals accumulating in the pipestems.

According to Bigham et al. (1994) and Schwertmann et al. (1995), jarosite forms under the most acidic conditions and at the highest sulfate concentrations (pH 1.5–3, [SO₄] >3000 mg l⁻¹) followed by schwertmannite (pH 3–4, [SO₄] = 1000–3000 mg l⁻¹) and goethite (pH <6, [SO₄] <1000 mg l⁻¹). Provided that the results of water extraction (1:5) extrapolated to a narrower soil-to-solution ratio gives an estimate of pore water composition, the 130–190 cm layer of the Ylistaro soil has favourable conditions for the formation of schwertmannite and the 170–190 cm layer probably also for jarosite. All other soil layers in Ylistaro and all investigated soil layers in the Tweed Heads soil had low SO₄-S concentrations conducive to the hydrolysis of these minerals. Elemental composition of the interior coating of the pipestem hole suggests that in the Tweed Heads soil that material likely consisted of almost pure goethite, showing that the pipestems had reached a mature stage resulting in the formation of, for example, goethite pseudomorphs after schwertmannite.

According to the morphological description (Yli-Halla, 1997), the Ylistaro soil contained pale yellow colours on prism faces, indicating the presence of jarosite. This mineral was scarce in the SEM images of the present study but it was clearly detected in the XRD analysis. Plenty of features with schwertmannite-like morphology were observed in the SEM but because it is a short-range order mineral, schwertmannite is not readily detected in XRD analysis. One hypothesis for the abundance of schwertmannite-like accumulations is that the rapid oxidation of sulfidic material induced by artificial drainage surpasses the dissolution of K-containing primary minerals so that there may not be enough K available for the abundant formation of jarosite.

Figure 6 indicates the two likely end stages of pipestem development in AS soil. Figure 6a,b (a cross-section of the root and surrounding matrix shown in Figure 6a in the capillary fringe) shows the initial stage of having a living plant root transporting O₂ into the surrounding saturated and reduced soil matrix in both the capillary fringe and beneath the water table. This O₂ initiates the oxidation of sulfides in the zone immediately surrounding the root. Figure 6c,d shows the final stages of O₂ diffusing downwards through the vacant root hole into the capillary zone and the surrounding saturated and reduced soil matrix only in the capillary fringe where the oxidation of sulfides continues in the zone immediately surrounding the root. In this latter stage, the zone surrounding the root is severely acidified as demonstrated by the precipitation of both jarosite and schwertmannite forming impregnations and coatings around the root hole.

It is noteworthy that in both sites, the S concentration of the pipestem matrices decreased substantially

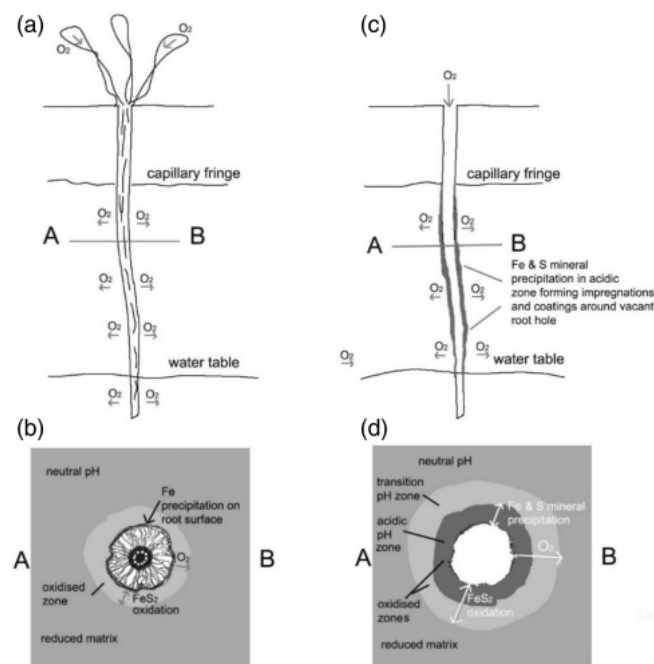


FIGURE 6 Schematic presentation of the formation of a pipestem in wetland soil starting from a living root in (a) and (b), and ending at a root hole after the death of the plant root in (c) and (d). In (a), section down the soil profile containing the living root is shown and (b) depicts a cross-section of the root and surrounding matrix across the plane indicated by A and B in (a). In (c) and (d), the corresponding diagrams are shown after the death and removal of the plant root.

downwards. In Ylistaro, this S decrease was associated with a major decrease of Fe while this trend was less pronounced in Tweed Heads where all pipestem matrices had a much higher Fe concentration than Ylistaro. We assume that the decreasing trend of S and Fe concentrations in Ylistaro pipestem matrices reflects the early development stage of these pipestems. The younger pipestems (i.e., those in the lower soil layers) had experienced less accumulation of Fe and S from the adjacent soil material and formation of the respective minerals than the older ones in the upper soil layers. It is also possible that a net upward transport of the mobile soluble $\text{SO}_4\text{-S}$ as a result of evapotranspiration may have increased the S concentration of the pipestems in the upper soil layers. In the Tweed Heads pipestems, the low S concentration, particularly compared to the high Fe concentration, suggests that the metastable Fe-S compounds had already hydrolysed to a considerable extent, resulting in the leaching of S while Fe mostly remains in the pipestems and has been converted to goethite. This degradation of the Tweed Heads pipestems seems to have taken place more intensively deeper in the soil profile. This assumption is supported by the fact that the infillings in the interior coatings were rather low in S but not

in Fe in the deeper horizons. It is these infillings that are most vulnerable to leaching and a likely decrease in S concentration.

5 | CONCLUSIONS

The pipestems and the desiccation cracks are essential in the ripening of the soil by allowing O_2 to enter the subsoil, resulting in the radial oxidation of the soil material surrounding the pore. Water soluble products of the oxidising reactions, $\text{SO}_4\text{-S}$ in the first place, can be transported out of the respective soil horizons during periods of water saturation and out of the soil through the drainage system of agricultural land. The presence of pipestems provides evidence for the plant root-induced oxidation of the subsoil well below the artificial drainage pipes regardless of whether the cylindrical pores formed by the plant roots contain living roots or are empty.

Even though there is uncertainty in the quantitative estimates of S mobilisation, they show the magnitude of S involved and redistributed in the soil profile as a result of pipestem formation. After exhaustion of the sulfidic material, S is still partly retained in the pipestem in the form of jarosite and schwertmannite, constituting a stock of retained acidity. Over time, these S-containing oxidation products are hydrolysed with associated leaching of S and acidity while Fe remains in the soil profile in the form of goethite. Comprehending the quantities of minerals involved in these processes helps us make realistic projections for the feasibility to manage the acid loading from AS soils once they have been allowed to oxidise.

AUTHOR CONTRIBUTIONS

Markku Yli-Halla: Investigation; funding acquisition; writing – original draft; conceptualization. **Leigh Sullivan:** Conceptualization; investigation; writing – review and editing; visualization; methodology; supervision; resources.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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